

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

VERILOG HDL AND PLD
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List the components of a VHDL module	M 1.04	R
2	Write any one data type used in VHDL	M 1.03	R
3	Write any two sequential models used in VHDL	M 2.01	R
4	Define User Defined Primitives(UDP)	M 2.02	R
5	Write how is inputs and outputs defined in a module in dataflow model	M 2.07	U
6	Name the model in VHDL used to describe a system whose digital logic structures are unknown	M 3.01	U
7	Name the sequential statement which appears inside process in VHDL	M 3.02	R
8	Write the use of logic block in CPLD architecture	M 4.02	U
9	Write down the different methods of programming PALs	M 4.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Describe the advantages of using HDLs over traditional schematic based design	M 1.01	U
2	Explain port declaration with an example using Verilog code	M 1.04	U
3	Explain gate level modelling of a half adder circuit	M 2.03	A
4	Explain the structure of a stimulus block	M 1.03	U
5	Write the syntax of forever loop in behavioral modelling	M 3.03	R
6	Write any 3 applications of PLA	M 4.01	U

7	Compare FPGA with CPLD	M 4.05	U
8	Differentiate between PAL and PLA	M 4.01	U
9	Write short note on PAL architecture	M 4.01	U
10	Explain how to interface ports in HDL	M 1.04	U

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	With a neat diagram explain the components of a Verilog module	M 1.04	U
OR			
2	Why VHDL has evolved as popular HDL in digital circuit design	M 1.01	U
3	Explain gate level modelling with any combinational circuit	M 2.03	U
OR			
4	Explain data flow modelling of ripple carry counter	M 2.08	U
5	Write VHDL for two 4 bit comparator using gate level modelling	M 2.03	A
OR			
6	Write VHDL code for 4X1 mux using data flow modelling	M 2.08	A
7	Explain about blocking assignments in behavioral modelling	M 3.02	U
OR			
8	Explain how delay control is done in behavioral modelling	M 3.01	U
9	Explain about sequential blocks in behavioral modelling	M 3.02	U
OR			
10	Write the VHDL description of AND gate in behavioral modelling	M 3.04	U
11	Explain various features of FPGA architecture	M 4.03	U
OR			
12	Write short note on CPLD architecture issues	M 4.02	U

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
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VERILOG HDL AND PLD
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define test bench in VHDL	M 1.03	R
2	Define modules in VHDL	M 1.04	R
3	Write any two parts of a simulation	M 1.03	U
4	List any two operator types used in data flow modelling	M 2.07	U
5	Write an example for conditional statement used in behavioural modelling	M 3.02	U
6	Write the use of always in behavioural modelling	M 3.01	R
7	Differentiate between identifiers and keyword with an example	M 1.03	U
8	Name two application areas of CPLD architecture	M 4.02	U
9	Write the two modes of programming FPGA	M 4.03	U

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Explain bottom-up design methodology	M 1.02	U
2	Explain lexical convention white space with examples	M 1.03	U
3	Write the verilog description of D latch using behavioural modelling	M 3.05	U
4	A full subtractor has three 1 bit inputs x,y,z(previous borrow) and two 1 bit outputs Difference(D) and borrow (B). Write verilog description using data flow modelling	M 2.08	A
5	Explain about non blocking assignments	M 3.01	U
6	Write short note on system tasks	M 1.03	U

7	Explain gate level modelling of T Flip Flop	M 2.03	U
8	Write any three applications of FPGA	M 4.03	U
9	Define PAL	M 4.01	R
10	Write short note on PLA architecture	M 4.01	R

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain top down design methodology with an example	M 1.02	U
OR			
2	Write short notes on trends in HDLs	M 1.01	U
3	Write verilog description of SR latch in gate level modelling	M 2.03	U
OR			
4	Write verilog description of 1:4 demultiplexer in data flow modelling	M 2.08	U
5	Design 4:1 multiplexer using gate level modelling	M 2.03	A
OR			
6	Write VHDL data flow description of JK Flip flop in dataflow modelling	M 2.08	A
7	Explain about task and how to declare a task	M 3.01	U
OR			
8	Explain how event control is done in behavioural modelling	M 3.01	U
9	Write the VHDL description of OR gate in behavioural modelling	M 3.04	U
OR			
10	Write the VHDL description of 4 bit counter using behavioural modelling	M 3.05	A
11	Explain about FPGA architecture with diagram	M 4.03	U
OR			
12	Explain PAL architecture with diagram	M 4.01	U